

Course Name: Probability Theory and Random Processes

Course Code: MA106

L-T-P scheme: 3-1-0

Credits: 4

Course Coordinator: Dr. D.K. Sharma

Prerequisite: Students should have the knowledge of set theory, statistics, differential & integral calculus.

Objective: Objective of this course is to provide a foundation in the theory and applications of probability and stochastic processes and an understanding of the mathematical techniques relating to random processes in the areas of signal processing, detection, estimation, and communication. Topics include the axioms of probability, random variables, and distribution functions; functions and sequences of random variables; stochastic processes; and representations of random processes.

Course Outcomes:

At the end of the course student will be able to:

		Blooms Level
CO1	Explain the sets, probability and conditional probability. State Bayes' theorem.	BL2
CO2	Describe random variables, probability function, probability density function and cumulative distribution function.	BL2
CO3	Apply probability distributions and correlation measures of bivariate random variables to solve the problems.	BL3
CO4	Analyze the stationarity of random process and its applications.	BL4
CO5	Examine the concept of reliability, hazard rate function, mean time to failure.	BL4

Course outline:

- Unit-1** Random experiments, sample space and events. Three basic approaches to probability, conditional probability, total probability theorem, Bayes' theorem of probability of causes, Bayes' theorem of future events, total independence, mutual independence, and pair wise independence.
- Unit-2** One-dimensional random variables (discrete and continuous) and their distributions, bivariate distributions, joint, marginal and conditional distributions, characteristic function.
- Unit-3** Covariance and correlation of random variables. Some special probability distributions: Binomial and Poisson probability distributions. Negative Binomial, Geometric, and Normal probability distributions. Fitting of probability distributions.
- Unit-4** Concept of reliability: Reliability function, hazard rate function, mean time to failure (MTTF), cumulative and average failure rate, conditional reliability and failure rates, residual MTTF, special failure rate distributions (Exponential and Weibull distributions), reliability of systems—series configuration and some deductions, parallel-series configuration, and series-parallel configuration.
- Unit-5** Introduction and description of random processes, average values of random processes, stationary processes and computation of their averages, autocorrelation function and its properties, cross-correlation and its properties.

Course Conduction Plan:

Topics	Lecture Hours
UNIT I	
Random experiments, sample space and events.	3
Three basic approaches to probability.	1
Conditional probability, Total probability theorem	2
Bayes' theorem of probability of causes, Bayes' theorem of future events	1
Total Independence	1
Mutual independence and pair wise independence.	2
UNIT II	
One dimensional random variables (discrete and continuous) and their distributions.	2

Bivariate distributions.	1
Joint and Marginal distributions.	2
Conditional distributions.	1
Characteristic function.	2
UNIT III	
Covariance and correlation of random variables.	3
Some special probability distributions: Binomial, Poisson, probability distributions. Negative Binomial.	5
Geometric and Normal probability distributions. Fitting of probability distributions.	2
UNIT IV	
Concept of reliability: Reliability function, Hazard rate function, Mean time to failure, Cumulative and average failure rate, Conditional reliability and failure rates, Residual MTTF.	1
Some special failure rate distributions- the exponential distribution and the Weibull distribution.	4
Reliability of systems- series configuration and some deductions, parallel-series configuration, series -parallel configuration.	2
UNIT V	
Introduction and description of random processes, average values of random processes.	4
Stationary processes and computation of their averages, autocorrelation function and its properties, Cross correlation and its properties.	2
Power spectral density function and its properties.	3
Ergodicity of a random process, Poisson processes	2

Methodology:

The course will be covered through lectures supported by tutorials. Apart from the discussions on the topics covered in the lectures assignments/ quizzes in the form of questions will also be given.

Evaluation Scheme

Evaluations	Marks	Remarks
T1	15 Marks (1 Hour)	UNIT1 & UNIT2
T2	25 Marks (1.5 Hours)	UNIT3 and around 30% from coverage of Test-1
T3	35 Marks(2 Hours)	UNIT4-6 and around 30% from coverage of Test-2
Assignments	10 Marks	4 or 5 Assignments to be given
Quiz	5 Marks	2 or 3 quizzes
Tutorials	5 Marks	
Attendance	5 Marks	
Total	100 Marks	

Learning Resources:

Tutorials and lecture slides on Probability Theory and Random Processes (will be added from time to time): Digital copy will be available on the JUET server.

Text books:

- [1] T. Veerarajan ,Probability, Statistics and Random Processes, Tata McGraw Hill.
- [2] J.J. Aunon & V. Chandrasekhar, Introduction to Probability and Random Processes, McGraw Hill International Ed.
- [3] A. Papoulis & S.U. Pillai, Probability, Random Variables and Stochastic Processes, McGraw Hill.
- [4] H. Stark, and J.M. Woods, Probability and Random Processes with Applications to Signal Processing, Pearson Education.

Related Online courses

Probability and Random Processes

https://onlinecourses.nptel.ac.in/e-learning/preview/noc22_ee123